

Soumendu Nandi

Solutions Architect | Generative AI | Azure | Data Product Engineering

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Leading as a Solutions Architect specializing in Generative AI, Data Product, and Azure Cloud, delivering scalable, secure, and governed cloud-native solutions. Leverage rich experience of architecture, AI product engineering, DevOps, LLMOps and Python/.NET expertise to drive innovation, modernization, and measurable business value.

Profile Summary

- Career & Architecture:** Solutions Architect with **nearly 18 years** of experience delivering cloud-native, AI-powered, and enterprise solutions across Life Sciences, Banking, Logistics, Manufacturing, Retail, and Title Insurance.
- Cloud & Technology:** Expertise in Azure Cloud, Generative AI, Data Engineering, Python/.NET, Azure OpenAI, FastAPI, LangChain, LangSmith, Databricks, Microsoft Fabric, and modern cloud architectures.
- Generative AI:** Architected Generative AI and Conversational AI solutions, including RAG and LLM-based applications, achieving 5 FTE annual effort reduction in Pharma Supply Chain Regulatory report generation. Get ready with Go To Market strategic platform pieces, components and reusable libraries, for Agent
- Data & Modernization:** Proven success in legacy modernization and data platform transformation, improving scalability, data accessibility, analytics, and operational efficiency.
- Architecture & Governance:** Strong expertise in Architecture Governance, DevSecOps, Security, and Compliance, with reusable patterns, CI/CD, Terraform, Docker, Kubernetes, and automated security scanning.
- Cloud-Native Architecture:** Experienced in microservices, event-driven, serverless, APIs, messaging, and integration architectures, with strong Authentication/Authorisation expertise, and Practising OSWAP - security best practise.
- Leadership & Stakeholders:** Proven technical leadership and stakeholder management, guiding cross-functional teams and aligning architecture with business, Enterprise Architecture, security, and FinOps objectives.
- Innovation & Optimization:** Passionate about AI adoption, scalable architecture, innovation, and cost optimization, with expertise in RAG, LangChain, PromptFlow, vector databases, MCP, A2A, and AI-assisted engineering.

Core Competencies

Strategy & Architecture

- Solution Architecture & Strategic Planning
- Enterprise Architecture Governance
- Generative AI Product Strategy
- Cloud Transformation & Adoption

Data & Digital Transformation

- Data Architecture & Modernization
- Digital Transformation
- Enterprise Integration Strategy
- Data Governance & Compliance

Governance & Delivery

- Solution Design & Delivery
- AI/LLM Ops Solution Governance
- Security & Risk Governance
- DevSecOps Strategy & Delivery

Leadership & Business

- Stakeholder & Executive Management
- Cross-Functional Team Leadership
- Technical Mentoring & Capability Development
- Continuous Improvement & Cost Optimization

Technical Skills

Cloud: Microsoft Azure, Azure PaaS, App Service, Container Apps, Functions, Logic Apps, APIM, ADF, Key Vault, AKS; GCP, Cloud Run, API Gateway, BigQuery, GCP Storage.

Generative AI & AI/ML: Azure OpenAI, Azure AI Search, Azure ML, AI Language, Azure Document Intelligence, Azure Open AI, Conversational AI, RAG, LLMs, NLP, LangChain, LangGraph, LangGraph SDK, LangSmith, PromptFlow, MCP, A2A, RPC, AI Chatbots, LLM Fine-Tuning & Evaluation, AI Responsibility, Streamlit, Vector Database - Chroma.

Data & Analytics: Microsoft Fabric, Azure Databricks, ADLS, ADF, PySpark, Data Engineering, DataOps, Data Governance, Data Analytics, Data Pipelines, BigQuery.

Programming: Python, C#, SQL, TypeScript, PowerShell, Bash, YAML.

Backend & APIs: .NET Core, ASP.NET MVC/Web API, Entity Framework, FastAPI, Flask, SQLAlchemy, REST APIs, WebSockets, RPC.

Frontend & Testing: React, Vite, Next.js, Angular, TypeScript, Vitest, Jest, Playwright.

DevOps & Infrastructure: Azure DevOps, Git, GitHub Actions, GitOps, Docker, Kubernetes, Terraform, ARM Templates, CI/CD, IaC, Build/Test Automation, SAST, SCA, Dependency & Container Scanning.

Databases & Storage: Azure SQL, SQL Server, Cosmos DB, SQLite, PostgreSQL/PGVector, Chroma, BigQuery, Azure/GCP Storage, Elasticsearch.

Architecture & Integration: Microservices, Event-Driven, Serverless, REST, API Gateway, Enterprise Integration, Data Streaming, Azure Service Bus, RabbitMQ, Kong API Gateway, Messaging Patterns.

Security & Identity: OAuth 2.0, Microsoft Entra ID/Azure AD, OpenID Connect, Okta, AWS Cognito, Authentication, Authorization, Security Governance, Compliance.

Monitoring & Delivery: Azure Monitor, Application Insights, Log Analytics, Prometheus, Grafana, Blue/Green & Rolling Deployments, Cloud-Native Development, DevSecOps, FinOps, Architecture & Solution Governance.

Certifications

- **Microsoft Certified: Azure Solutions Architect Expert (AZ-305)** — Microsoft
- **Microsoft Certified: Azure Architect Technologies (AZ-303)** — Microsoft
- **Microsoft Certified: Azure Developer Associate (AZ-204)** — Microsoft
- **Microsoft Certified: Azure AI Fundamentals (AI-900)** — Microsoft
- **Microsoft Certified: Azure Fundamentals (AZ-900)** — Microsoft

Work Experience

Architect | Cognizant | Kolkata, India | May 2019 – Present

Responsibilities:

- Spearheading end-to-end architecture for AI/ML and cloud-native solutions from inception through deployment and hypercare, ensuring scalability, performance, security, and compliance.
- Governing enterprise architecture standards and aligning technical decisions with business, Enterprise Architecture, and FinOps objectives.
- Defining reusable architecture blueprints, integration patterns, and GenAI frameworks to improve consistency and accelerate delivery.
- Architecting Generative AI and Conversational AI solutions using Azure OpenAI, Python, FastAPI, RAG, prompt flows, LangChain, LangGraph, LangGraph SDK, Chroma, and PGVector.
- Driving DevSecOps and infrastructure governance through GitOps, IaC, CI/CD, SAST, SCA, container scanning, and automated security controls.
- Leading data platform modernization using Azure, Databricks, Microsoft Fabric, and DataOps to enable scalable analytics and enterprise data access.
- Heading cross-functional teams and security architecture, leveraging Entra ID/Azure AD and OAuth2 while promoting AI-assisted engineering.

Achievements:

- Delivered Generative AI modules for Pharma Supply Chain, reducing annual effort by 10–15 FTEs through automation.
- Modernized a legacy data platform for a global manufacturer, improving scalability, accessibility, and analytics.
- Established reusable architecture and CI/CD security patterns, accelerating delivery and strengthening compliance.
- Drove AI-assisted development and team capability building, improving productivity through mentoring and training.
- Designed multi-tenant AI architecture for scalability and cost optimization; recognized for Architecture Governance excellence.

Technical Architect / Tech Lead | Cognizant – Generative AI Solution, Pharma Supply Chain Domain | Kolkata, India Oct 2023 – Mar 2026

Responsibilities:

- Led Generative AI and Conversational AI development, including prompt flows and enterprise AI capabilities.
- Designed an AI-driven document automation platform supporting healthcare/pharma compliance and governance.
- Architected data-lake integrations using PySpark, ADLS, and Azure Databricks for AI workflow.
- Designed scalable back-end communication using REST APIs, WebSockets, and RPC, defining the cloud deployment strategy.
- Led stakeholder workshops to gather requirements, structure solutions, and align delivery with business goals.
- Established reusable CI/CD and security patterns and championed GitHub Copilot adoption to improve engineering productivity.

Achievements:

- Delivered multiple production-oriented Generative AI capabilities using Azure OpenAI, Python, FastAPI, LangChain, and Streamlit, supporting business process automation.
- Strengthened protection of sensitive healthcare data through robust authentication and authorization architecture aligned with regulatory requirements.
- Improved deployment readiness and security posture by embedding automated security controls directly into the development lifecycle.
- Optimized data-processing workflow to support scalable AI workloads and more efficient critical operations.

Technical Architect / Tech Lead | Cognizant – Partner Compensation, Logistics Domain | Kolkata, India | May 2022 – Sep 2023

Responsibilities

- Partnered with clients from project inception through hypercare, providing cloud architecture, application architecture, API development, and CI/CD leadership.
- Led technical development across multiple modules and coordinated end-to-end solution integration.
- Designed reusable architecture patterns and deployment pipelines to support scalable and repeatable delivery.
- Worked with cross-functional teams on solution structuring, project execution, and data governance.
- Integrated application and infrastructure security checks into CI/CD workflow, including code, dependency, SAST, SCA, and container scanning.

Achievements:

- Streamlined API integration & delivery processes, improving solution responsiveness & reducing integration complexity.
- Established reusable delivery patterns that supported faster deployment and scalable solution implementation.
- Strengthened data governance practices to improve data integrity and regulatory alignment.

Tech Lead / Technical Architect | Cognizant – Retail Fast Service Restaurant, Inventory Management / Back of House | Kolkata, India

Jun 2019 – Apr 2022

Responsibilities

- Led the technical architecture and development of inventory management and Back-of-House solutions.
- Managed development teams throughout the project lifecycle, providing technical direction, mentoring, and delivery oversight.
- Designed integrations across .NET/Web APIs, Azure services, databases, & mobile applications to support end-to-end business processes.
- Directed solution structuring and technical decision-making while ensuring integration with existing enterprise systems.
- Developed training materials and conducted workshops to support adoption of newly implemented inventory-management capabilities.

Achievements:

- Implemented real-time inventory tracking capabilities to improve inventory visibility and operational efficiency.
- Enhanced integration between inventory solutions and existing business systems, supporting more streamlined operational workflow.
- Improved user adoption through structured training, knowledge transfer, and stakeholder engagement.

Technical Lead / Senior Software Engineer | FNF India Pvt. Ltd. | Bangalore, India | May 2016 – Apr 2019

Senior Software Engineer | Infinite Computer Solutions | Bangalore, India | May 2015 – Apr 2016

Software Developer | Nexval Infotech | Kolkata, India | Oct 2014 – May 2015

Software Developer / Support Personnel | National Informatics Centre (NIC) | Kolkata, India | Dec 2008 – Sep 2014

Education

Master of Computer Applications (MCA) — West Bengal University of Technology, West Bengal | 2008

B.Sc. – Mathematics, Physics & Chemistry — University of North Bengal, West Bengal, India | 2005

Personal Details

Languages Known: English, Bengali, Hindi | **Address:** Kolkata – 700156, West Bengal | **Year of Birth:** 1983